

# Nosecone-Whisker Aerodynamics

## Tire-Wake & Front-Wing Flow Control: Whisker Placement and Design Notes

### 1. Introduction

In terms of nosecone design this year, we have a few extra devices and features we want to try to carry forward this year. Our new design this year will implement a new whisker device that generates a small vortex using winglets mounted near the nosecone or front suspension. This paper covers how each placement affects the front tire wake and front wing flow as well as what still needs validating before we lock down the design for this year.

### 2. What are these whiskers?

In terms of an Aerodynamics package, the whiskers are a small pair of upright airfoils that run with or without endplates and are positioned so that a vortex is generated to reshape the air the moves through it. Unlike a normal wing, it is not an Aerodynamic device and thus generates no downforce. Anything that is technically part of the nosecone may not be an aerodynamic device. Inspiration from this was taken from UoFT and ETS,

Definitions for understanding purposes:

#### **Shutters:**

Shutters are often mistaken for whiskers and vice versa but they're completely different. Imagine you're about to go to bed and before going to sleep, the glaring light of the moon stares upon your resting eyes and you realize you can't go to sleep. You realize that you have to close the blinds to lock out that pestering glow and so you go and close your **shutters**. Now imagine those shutters are now on the sides of the car straightening out the airflow that passes through it. That is what the shutters are. They are placed strategically where turbulent air threatens undertray and wing performance. The reason they are necessary is because air coming in contact with front tires generate a highly chaotic wake filled with low-pressure vortices which hits downstream aero surfaces like inlets and wide wings dropping downforce immensely. They do this through vorticity reduction as the air is straightened into aligned parallel streams when the air hits the angled blades, forcing the air onto the downstream surfaces preventing flow separation on the back end of the car.

### 3. The Problem: Tire Wake + Front-Wing Upwash

As the nosecone is not an aerodynamic device, we must shift our design to focus on supporting other subsystems in hopes of enhancing their efficiency and performance. One focus for us is integrating our nosecone with front wing designs to generate that downforce that it needs through up washing while keeping the

downstream air clean as well. When the front tire's wake and the turbulence that come from suspension linkages come together, it collapses into a wide array of low pressure, high-vorticity air taking up right where the undertray and rear wing need clean air to perform. Sources online and other teams suggest this effect is quite significant in playing a role in the entire Aero package's efficiency.

## 4. How Whiskers Fixes It

The Vortex generated at the wingtip of the Whisker does two things to incoming upwash that passes around it. First off, the whisker imparts a downward kinetic energy pushing the upwash back down instead of letting it rise. Secondly its outwashes the air, spreading it laterally and away from the centerline. To explain it further, it basically uses its own vortex rotation flings the upwash back down. The whisker is placed at angle where the disturbed air is, and the vortex pushes the air downwards and sideways away at the same time to create that outwash. As a result, less turbulent air is being dumped into the middle/rear end of the car carrying benefits to the rear wing as well.

However, that benefit can be costly as the whisker generates small positive lift that can oppose downforce. It may reduce front end downforce on straights for a cleaner streamline downstream. That's not necessarily game ending but its more of a lever that can be used to balance the center of pressure.

## 5. Placement: location vs performance

| Position                         | What you get                                                          | Good For                                                                |
|----------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| Above the nosecone               | Largest wingtip vortex which generates the strongest vortex           | Maximum downward/outward push and clean, unobstructed span to work with |
| In front of the front suspension | Applies the vortex before the airstream hits the suspension linkages. | Suspension wake is a major contributor to disturbed-flow region         |
| Above the front suspension       | Best support for suspension wake and maximum outwash                  | Probably the most reasonably good all-around solution                   |

The benefits for each are circumstantial and we may be able to test positioning in the future. Nosecone mounting is good because it gives us more whiskers span to work with and works with cleaner air which is why it produces the strongest downwards and outwards vortex. Mounting adjacent to the suspension honestly is more precise as it works right where the suspension linkages create the messy air. Ultimately when it comes to choosing the optimal placement is finding whether upwash spread or suspension turbulence is more impactful overall using our CFD sims tufts in windtunnel.

## 6. Design Parameters Worth Tuning

**Fin shape:**

**Angle of incidence:** We used -10 degrees which was common with what other teams were using for theirs, positioning and span may affect this.

**Element count:** 1

**Rule envelope:** Compliant with FSAE Section VE.1 / T.7 limits, staying forward of the front axle centerline, maintaining a span within 400mm of chassis centerline, keeping overall height below the 250mm wheel zone threshold (or inboard of the inner tire face), and adhering to mandatory 5mm / 3mm leading-edge radii safety requirements.

## 7. Whole-Vehicle Balance

Adding the whiskers can cause lift which shifts the center of pressure. In some studies, optimizing shutters on its own can increase the forward pressure that could result in a handling problem with the car. The fix to that front end gain was to place a whisker to generate lift at the front. This is an important criteria we need to consider when looking at constraints.

To measure this, whenever we model or test a whisker, we will try to consider the center of pressure and local wake cleanup. Controlling the inflow but messing up aero balance everywhere is not a win until we can find a way to compensate.

## 8. Check List

- Ran a CFD sweep of angle of incidence and mounting position matched to your actual front wing's upwash. A whisker sized for someone else's front wing won't transfer directly onto yours.
- (Eventually) Validate across cornering yaw range and not just zero-yaw straight-line.
- Check the whole-vehicle center-of-pressure and pitching-moment effect alongside the local wake benefit
- Confirm final geometry (width, height, ground clearance) against 2027 aerodynamics envelope before committing to a mounting position.
- Decide the primary objective (largest vortex vs. protection vs. balanced performance) before picking a position.

## References

Qi, X., Ou, Y., Zhang, H., & Wang, D. (2022). Efficiency Enhancement Design Approach in the Side Wing of a FSAE Car Utilizing a Shutter-Like Fairing Structure. *Applied Sciences*, 12(13), 6552. <https://doi.org/10.3390/app12136552> (open access)